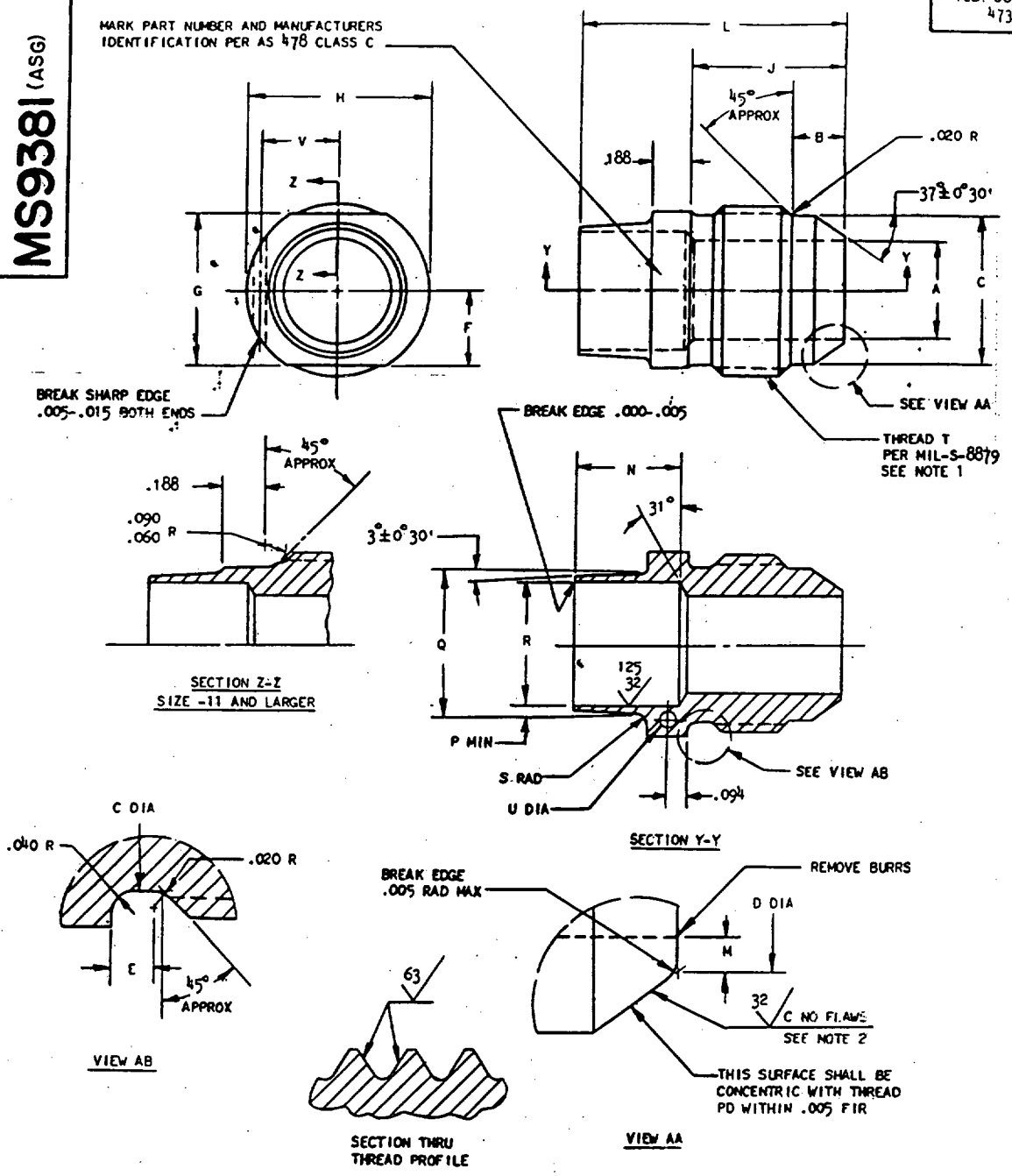
MS9381 (ASG)

FED. SUP CLASS 4730




1. THREAD ON SIZES -16 AND SMALLER SHALL BE ROLLED PER AMS7472.
2. CONICAL SURFACE SHALL BE FREE OF NICKS, CHATTER MARKS OR LOW SPOTS. UNBROKEN LINE OF CONTACT TO SHOW ALL AROUND AT OTHER THAN EXTREMITIES OF SURFACE WHEN ROTATED IN CONTACT WITH 74° 0' ±0° 1' INCLUDED ANGLE GAGE LIGHTLY COATED BY RUBBING WITH CARBON PAPER.
3. ALL DIAMETERS SHALL BE CONCENTRIC WITHIN .010 FIR UNLESS OTHERWISE SPECIFIED.
4. DIMENSIONS ARE TO SHARP CORNERS UNLESS OTHERWISE SPECIFIED.
5. MATERIAL: CORROSION RESISTANT STEEL, AMS 5646.
6. CLEANING: FINISHED PARTS SHALL BE DEGREASED AND IMMERSED FOR NOT LESS THAN 20 MINUTES IN A SOLUTION OF 1 VOLUME OF NITRIC ACID (SP GR 1.42) AND 9 VOLUMES OF WATER AT ROOM TEMPERATURE.
7. SURFACE ROUGHNESS: AS 291. UNLESS OTHERWISE SPECIFIED SURFACES TO BE 125 MICROINCHES.
8. BREAK SHARP EDGES .003-.015, UNLESS OTHERWISE SPECIFIED.
9. DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: LINEAR DIMENSIONS ±.010, ANGULAR DIMENSIONS ±5°.
10. DO NOT USE UNASSIGNED PART NUMBERS.

AS & AMS ARE SOCIETY OF AUTOMOTIVE ENGINEERS, INC. PUBLICATIONS.
THIS STANDARD WAS DEVELOPED COOPERATIVELY WITH THE SAE AEROSPACE PROPULSION DIVISION.

Review activities: Navy - AS, USAF - 11

This military standard is approved by the Department of the Air Force and the Naval Air Systems Command and is mandatory for use by these activities. All other military activities are required to employ this standard where suitable.

APPROVED 3 Oct 66 REVISED

| | | |
|---|---|---|
| P.A. USAF - 11<br>Other Cust Navy - AS | TITLE<br>COUPLING, TUBE - BRAZING, AMS 5646 | MILITARY STANDARD<br>MS9381 (ASG) |
| PROCUREMENT SPECIFICATION NONE | SUPERSEDES: | SHEET 1 OF 2 |

DD FORM 672-1 (Limited coordination) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE

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and the Naval Air Systems Command and is mandatory for use by these activities. All other military activities are required to employ this standard where suitable.

Navy - AS
USAF - 11

# MS 9381 (ASG)

| PART NO. | NOM TUBE OD | THREAD T | | | | | A | B +.015 -.000 | C DIA +.000 -.010 | D DIA +.003 | E +.015 -.000 | F +.005 | G +.005 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | SIZE | MAJOR DIA | PITCH DIA | MINOR DIA | ROOT RAD | | | | | | | |
| MS9381-02 | .125 | .3125-24UNJF-3A | .3053- .3125 | .2827- .2854 | .2591- .2644 | .0063-.0075 | .058- .065 | .144 | .245 | .132 | .063 | .125 | .250 |
| MS9381-03 | .188 | .375-24 UNJF-3A | .3678- .3750 | .3450- .3479 | .3214- .3268 | .0063-.0075 | .121- .128 | .144 | .307 | .195 | .063 | .156 | .312 |
| MS9381-04 | .250 | .4375-20UNJF-3A | .4294- .4375 | .4019- .4050 | .3736- .3797 | .0075-.0090 | .168- .175 | .160 | .359 | .242 | .075 | .188 | .375 |
| MS9381-05 | .312 | .500-20UNJF-3A | .4919- .5000 | .4643- .4675 | .4360- .4422 | .0075-.0090 | .230-.237 | .160 | .421 | .304 | .075 | .219 | .438 |
| MS9381-06 | .375 | .5625-18UNJF-3A | .5538- .5625 | .5230- .5264 | .4916- .4983 | .0083-.0100 | .293-.301 | .165 | .476 | .368 | .083 | .250 | .500 |
| MS9381-07 | .438 | .625-18UNJF-3A | .6163- .6250 | .5854- .5889 | .5540- .5608 | .0083-.0100 | .355-.363 | .165 | .539 | .430 | .083 | .281 | .562 |
| MS9381-08 | .500 | .750-16UNJF-3A | .7406- .7500 | .7056- .7094 | .6702- .6778 | .0094-.0113 | .387-.395 | .229 | .654 | .462 | .094 | .344 | .688 |
| MS9381-09 | .562 | .8125-16UNJ-3A | .8031- .8125 | .7683- .7719 | .7329- .7403 | .0094-.0113 | .434-.442 | .229 | .716 | .509 | .094 | .375 | .750 |
| MS9381-10 | .625 | .875-14UNJF-3A | .8647- .8750 | .8245- .8286 | .7841- .7925 | .0107-.0129 | .480-.488 | .255 | .767 | .555 | .107 | .391 | .781 |
| MS9381-11 | .688 | 1.000-12UNJF-3A | .9886-1.0000 | .9415- .9459 | .8944- .9038 | .0125-.0150 | .542-.552 | .304 | .875 | .619 | .125 | .438 | .875 |
| MS9381-12 | .750 | 1.0625-12UNJ-3A | 1.0511-1.0625 | 1.0042-1.0084 | .9570- .9663 | .0125-.0150 | .604-.614 | .304 | .938 | .681 | .125 | .469 | .938 |
| MS9381-14 | .875 | 1.1875-12UNJ-3A | 1.1761-1.1875 | 1.1291-1.1334 | 1.0820-1.0913 | .0125-.0150 | .729-.739 | .304 | 1.062 | .806 | .125 | .531 | 1.062 |
| MS9381-16 | 1.000 | 1.3125-12UNJ-3A | 1.3011-1.3125 | 1.2541-1.2584 | 1.2070-1.2163 | .0125-.0150 | .839- .851 | .312 | 1.188 | .918 | .125 | .594 | 1.188 |
| MS9381-18 | 1.125 | 1.500-12UNJF-3A | 1.4886-1.5000 | 1.4411-1.4459 | 1.3940-1.4038 | .0125-.0150 | .948- .960 | .312 | 1.375 | 1.027 | .125 | .656 | 1.312 |
| MS9381-20 | 1.250 | 1.625-12UNJ-3A | 1.6136-1.6250 | 1.5665-1.5709 | 1.5193-1.5288 | .0125-.0150 | 1.073-1.086 | .363 | 1.501 | 1.153 | .125 | .719 | 1.438 |
| MS9381-24 | 1.500 | 1.875-12UNJ-3A | 1.8636-1.8750 | 1.8164-1.8209 | 1.7692-1.7788 | .0125-.0150 | 1.307-1.320 | .374 | 1.750 | 1.387 | .125 | .844 | 1.688 |
| MS9381-28 | 1.750 | 2.250-12UNJ-3A | 2.2386-2.2500 | 2.1914-2.1959 | 2.1442-2.1538 | .0125-.0150 | 1.542-1.557 | .447 | 2.125 | 1.652 | .125 | 1 000 | 2.000 |
| MS9381-32 | 2.000 | 2.500-12UNJ-3A | 2.4886-2.5000 | 2.4413-2.4459 | 2.3941-2.4038 | .0125-.0150 | 1.776-1.791 | .457 | 2.375 | 1.886 | .125 | 1.125 | 2.250 |
| MS9381-40 | 2.500 | 3.000-12UNJ-3A | 2.9886-3.0000 | 2.9412-2.9459 | 2.8940-2.9038 | .0125-.0150 | 2.276-2.291 | .391 | 2.882 | 2.386 | .125 | 1.375 | 2.750 |
| MS9381-48 | 3.000 | 3.500-12UNJ-3A | 3.4886-3.5000 | 3.4411-3.4459 | 3.3940-3.4038 | .0125-.0150 | 2.776-2.791 | .391 | 3.382 | 2.886 | .125 | 1.625 | 3.250 |

| PART NO. | H DIA +.010 -.000 | J | L | M MIN | N | P MIN | Q DIA +.010 -.000 | R DIA +.002 -.000 | S | U | V | APPROX WEIGHT LB/100 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| MS9381-02 | .344 | .415 | .841 | .030 | .346 | .022 | .170 | .115 | .031-.062 | .045-.050 | .108-.112 | 1.17 |
| MS9381-03 | .438 | .446 | .884 | .030 | .358 | .023 | .234 | .178 | .031-.062 | .057-.062 | .139-.149 | 1.50 |
| MS9381-04 | .500 | .517 | .973 | .030 | .376 | .026 | .303 | .240 | .047-.078 | .057-.062 | .170-.180 | 2.05 |
| MS9381-05 | .593 | .517 | .973 | .030 | .376 | .029 | .370 | .302 | .047-.078 | .065-.075 | .207-.217 | 2.64 |
| MS9381-06 | .656 | .523 | .979 | .030 | .376 | .029 | .433 | .365 | .047-.078 | .065-.075 | .238-.258 | 3.12 |
| MS9381-07 | .719 | .523 | 1.006 | .030 | .403 | .033 | .504 | .428 | .047-.078 | .065-.075 | .270-.290 | 3.98 |
| MS9381-08 | .781 | .633 | 1.145 | .030 | .432 | .034 | .568 | .489 | .047-.078 | .065-.075 | .300-.320 | 5.82 |
| MS9381-09 | .844 | .633 | 1.164 | .030 | .451 | .038 | .637 | .551 | .047-.078 | .065-.075 | .331-.351 | 6.66 |
| MS9381-10 | .904 | .747 | 1.306 | .030 | .479 | .039 | .703 | .614 | .047-.078 | .065-.075 | .362-.382 | 8.41 |
| MS9381-11 | 1.000 | .853 | 1.443 | .030 | .510 | .041 | .769 | .677 | .047-.078 | .065-.075 | .410-.430 | 11.80 |
| MS9381-12 | 1.094 | .853 | 1.443 | .030 | .510 | .041 | .831 | .739 | .047-.078 | .065-.075 | .440-.460 | 13.13 |
| MS9381-14 | 1.188 | .898 | 1.488 | .030 | .510 | .041 | .956 | .864 | .047-.078 | .065-.075 | .503-.523 | 15.96 |
| MS9381-16 | 1.312 | .908 | 1.498 | .030 | .510 | .046 | 1.091 | .989 | .078-.109 | .065-.075 | .565-.585 | 18.27 |
| MS9381-18 | 1.500 | .954 | 1.544 | .030 | .510 | .046 | 1.216 | 1.114 | .078-.109 | .065-.075 | .644-.664 | 22.33 |
| MS9381-20 | 1.625 | .954 | 1.544 | .030 | .510 | .051 | 1.351 | 1.239 | .078-.109 | .065-.075 | .706-.726 | 27.38 |
| MS9381-24 | 1.875 | 1.079 | 1.669 | .030 | .510 | .061 | 1.616 | 1.484 | .078-.109 | .065-.075 | .830-.850 | 38.13 |
| MS9381-28 | 2.250 | 1.204 | 1.794 | .044 | .510 | .071 | 1.886 | 1.734 | .078-.109 | .065-.075 | .986-1.006 | 61.64 |
| MS9381-32 | 2.500 | 1.329 | 1.919 | .044 | .510 | .086 | 2.166 | 1.984 | .078-.109 | .065-.075 | 1.111-1.131 | 79.24 |
| MS9381-40 | 3.000 | 1.105 | 1.695 | .044 | .510 | .086 | 2.666 | 2.484 | .094-.125 | .065-.075 | 1.361-1.381 | 82.11 |
| MS9381-48 | 3.500 | 1.183 | 1.773 | .044 | .510 | .091 | 3.176 | 2.984 | .094-.125 | .065-.075 | 1.611-1.631 | 105.85 |

THIS STANDARD WAS DEVELOPED COOPERATIVELY WITH THE SAE AEROSPACE PROPULSION DIVISION.

APPROVED 3 Oct 66 REVISED

PROCUREMENT SPECIFICATION NOTE: P.A. USAF - 11; Other Cust Navy - AS

SUPERSEDES:

TITLE: COUPLING, TUBE - BRAZING, AMS 5646

MILITARY STANDARD

MS 9381 (ASG)

SHEET 2 OF 2

FED. SUP CLASS 4730

FORM 4721 (Limited modification)

PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE

U.S. GOVERNMENT PRINTING OFFICE: 1967-251-414/3951

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